KERALA STARTUP MISSION

RESEARCH INNOVATION NETWORK KERALA

Details of commercializable technologies onboarded to RINK from research institutions in Kerala

MAY 2023
Kerala Startup Mission

About Us

KERALA STARTUP MISSION

Kerala Startup Mission (KSUM) is the nodal agency for the promotion of technology-based innovation and entrepreneurship in the state of Kerala, India. KSUM has initiated a scheme titled 'Research Innovation Network-Kerala (RINK)', which connects research institutions with startups and industries to take research outcomes into market and develop commercial ventures.

KSUM had onboarded commercializable technologies from research institutions in Kerala. The details of the technologies are available in the booklet.

Technology Transfer & Commercialization Support Scheme

Kerala Government, through Kerala Startup Mission has launched a scheme called 'Technology Transfer & Commercialization Support Scheme', startups who have purchased or sourced technology license from any Government research Institution in India and are working on it to develop as a commercializable product. This scheme intends to support Technology License/Transfers to startups by providing a funding support of maximum of Rs.10 Lakhs in the form of reimbursement for the startups, who have paid to the Research Institution. However, the support is limited to 90% of the technology fee to be paid to the Research entity.

Contact Us

Kerala Startup Mission
G3B, Thejaswini, Technopark
Trivandrum, Kerala-695581
startupmission.kerala.gov.in

Research Innovation Network-Kerala (RINK)
E-mail: rink@startupmission.in
Website : https://rink.startupmission.in/

Contents

No	Name of Institution	Page No
1.	ICAR-Central Plantation Crops Research Institute (CPCRI) , Kasaragod	1
2.	ICAR-Central Tuber Crops Research Institute (CTCRI), Thiruvananthapuram	3
3.	CSIR–National Institute for Interdisciplinary Science and Technology (NIIST), Thiruvananthapuram	6
4.	Kerala University of Fisheries and Ocean Studies (KUFOS), Kochi	9
5.	ICAR- Sugarcane Breeding Institute Research Centre, Kannur	11
6.	Kerala Agricultural University, Vellanikkara	13
7.	ICAR- Indian Institute of Spices Research, Kozhikode	14
8.	Centre for Development of Advanced Computing, Thiruvananthapuram	16
9.	Center for Materials for Electronics Technology (C-MET), Thrissur	19
10.	Sree Chitra Tirunal Institute for Medical Sciences & Technology, Trivandrum	21

ICAR-Central Plantation Crops Research Institute
Kasaragod, Kerala-671124, INDIA

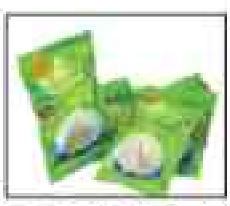
Cocosap chiller for the collection of hygienic and unfermented Neera/Kalparasa

Coconut sap chiller is a portable device developed by ICAR-CPCRI for the collection of hygienic and unfermented sap (Kalparasa®) from the coconut spadix. As per the demand of stakeholders, the institute has designed and fabricated molds required for the large-scale production of coconut sap chiller.

Virgin coconut oil (VCO)

ICAR CPCRI developed a complete package for producing the most pure form of Virgin Coconut Oil. We have two methods of VCO production, namely the Hot process and the fermentation or cold process. Used as functional, cosmetic and pharmaceutical oil.

Coconut Chips



Crispy, Crunchy, Sweet/Salty, delicious nutritionally rich chips prepared out of 8-9 months matured coconut kernels. ICAR-CPCRI has standardized the process technology for the production of different varieties of coconut chips.

Kalpa Organic Gold (coconut leaf vermicomposting technology)

Kalpa Organic Gold is granular organic manure produced by converting lignin-rich coconut leaves to vermicompost using an earthworm, *Eudrilus sp.* (African nightcrawler) isolated from CPCRI fields. The vermicompost can be applied to all field and horticultural crops.

Kalpa Soil Care (urea-free coir-pith composting technology)

Kalpa Soil Care is a simple technology for conversion of coir-pith to compost using organic inputs such as poultry manure, lime and rock phosphate. It is also suitable for all types of crops.

Kalpa Vermiwash (a liquid organic fertilizer from coconut leaf vermicompost)

KalpaVermiwash is a liquid organic fertilizer extracted from active vermicomposting of coconut leaves. The liquid fertilizer is to be diluted and applied to field and horticultural crops particularly vegetables.

KeraProbio® (a PGPR- based biofertilizer for coconut seedlings) KeraProbio® is a microbial biofertilizer. Bacillus megaterium is the rhizobacteria possessing multiple plant growth promoting properties that has been isolated from healthy palms. It is mixed in talc powder to make the formulation. Its application helps in raising healthy coconut seedlings.	**Coconut Embryo Culture** The ICAR-CPCRI EMBRYO CULTURE PROTOCOL permits safe germplasm exchange and helps in propagation of special types of coconut (soft and sweet endosperm types).
Coconut Pollen Cryopreservation Protocol developed at ICAR-CPCRI is simple and user friendly. Pollen collected from desiccated male flowers of high yielding palms can be retrieved and utilized in breeding programmes as and when required.	**Coconut Immature Inflorescence Tissue Culture** The immature inflorescence tissue culture protocol can produce uniform clones of elite palms.
Immature Inflorescence Tissue Culture In Arecanut This protocol enables mass multiplication of dwarf areca hybrids and YLD disease resistant palms.	**Molecular-Marker Based Confirmation of Coconut Hybrids** Microsatellite markers developed at ICAR-CPCRI assists in screening the progenies from the crosses of the parental palms to confirm their hybrid nature in the nursery itself. Use of these markers will ensure supply of genuine hybrid seedlings to farmers.
Kalpa EPN (CPCRI – SC1) Aqua formulation Technology This protocol enables mass multiplication of dwarf areca hybrids and YLD disease resistant palms.	***Trichoderma harzianum* (CPTD28) coir pith cake** Technology on Trichoderma coir pith cake production is simple and low- cost technology with a shelf life of one year. This formulation is very easy to handle and apply in the crowns of coconut palms as a prophylactic treatment against bud rot and in the management of stem canker of cocoa and also for the management of soil-borne diseases in potted plants.
Technology on mass production of Trichoderma harzianum (CPTD28) using areca leaf sheath Trichoderma multiplication using areca leaf sheath is a very simple, cost-effective technology for rapid mass production of *Trichoderma harzianum* (CPTD 28). Trichoderma enriched areca leaf sheath can be effectively utilized for eco-friendly management of stem bleeding and basal stem rot diseases of coconut and also for quick wilt disease of black pepper.	

ICAR-CENTRAL TUBER CROPS RESEARCH INSTITUTE

Amadi Nagar, Sreekariyam, Thiruvananthapuram, Kerala 695017

Fried snack foods and fried chips from Tapioca

Technologies included - Tapioca hot fries, tapioca hot sticks, tapioca pakkavada (salty fries), salty delight, tapioca murukku, tapioca crisps, tapioca nutrichips (with egg) and tapioca nutrichips (without egg) and fried chips from tapioca and sweet potato. Prepared using composite flour based on cassava and have high nutritional and textural quality & longer shelf life.

Functional Noodles from Cassava/ Sweet Potato

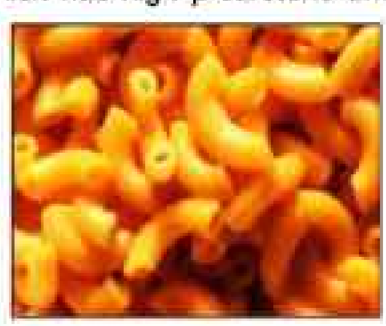
Functional noodles were prepared from sweet potato/ cassava starch by fortifying them with ingredients such as resistant starch and fibers. These noodles have a medium glycaemic index. Starch noodles developed from tuber starches such as cassava and Sweet potatoes are essentially gluten-free and hence an ideal choice for celiac patients. A source of protein 20% retention in cooked sweet potato noodles.

Functional Pasta from Cassava

The fortified pasta (whey protein, betanine, dietary fibre) prepared from cassava is gluten-free, has a medium glycemic index and provides 10-15% additional protein and fibre.

Orange Fleshed Sweet Potato Pasta

Orange-fleshed sweet potato is rich in β Carotene, a precursor of Vitamin A. High carotene content can help to combat Vitamin A deficiency. Pasta made from orange fleshed sweet potato which has high β-carotene and lysine content compared to wheat pasta, good swelling index after cooking, and a high amount of resistant Starch.



Cassava Resistant Starches – RS4 & RS5

Quick-cooking dehydrated cassava and Amorphophallus tubers were developed to reduce bulkiness during export and to increase the shelf life of the product with a cooking time of only 3-6 minutes, depending on the tubers.

Wax Coating Technology for Cassava Tubers

The surface coating of cassava roots is done by dipping them in hot/molten paraffin wax/other suitable wax and drying them on a bench before packaging them into boxes.

The wax treatment provides a barrier for the exchange of gases such as oxygen, carbon dioxide and water vapour and keeps the roots fresh for up to a month or two. This technology helps the processors and traders to store, transport and sell fresh cassava roots to distant places for an extended period of time and increased profitability.

Super Absorbent Polymer from Cassava Starch

Semi-synthetic super absorbent polymer (SAP) prepared from cassava starch. The addition of hydrogel to the soil and irrigating field at an interval of once in three days could increase the porosity and water holding capacity of the soil while maintaining higher nutrient levels for better plant growth. The SAP is a useful soil additive during drought and other moisture-stress conditions.

Micronutrient Foliar Formulations for Site-Specific Nutrient Management of Tropical Tuber Crops

Five micronutrient formulations were developed through soil inventorisation, analysis of soil and plant samples and Site-Specific Nutrient Management experiments and modeling studies conducted under different situations. They are

i. Cassava in acid soils (Zn, B)
ii. Cassava in neutral to alkaline soils (Fe, Mn, Zn, Cu)
iii. Elephant foot yam (Zn, B, Fe, Mn, Cu)
iv. Yams (Zn, B, Mn, Fe, Cu)
v. Sweet potato (Zn, B)

Electronic Crop (E Crop) The Electronic Crop (E Crop) is an IoT technology that simulates crop growth in response to weather and soil parameters and also generates an agro advisory that is sent to the farmer's mobile by SMS. Based on the climate and soil data, it calculates the requirements of fertilizers, water, agronomic and cultural practices to be followed to get more yield. Using sensors, the solar-powered device collects real-time data on the maximum and minimum temperature, solar radiation, relative humidity, precipitation, soil moisture content, wind velocity, and wind direction. The data are sent to a central server through a modem and run through a simulator to generate the advisory.	**SPOTCOMS** SPOTCOMS is a growth simulation model of sweet potato which predicts the crop phenology in response to environmental factors. It mimics the growth and development of sweet potato and helps to study crop growth and to calculate growth responses to the environment. This model is validated for nearly 30 Indian varieties and other varieties of Uganda, Indonesia, Vietnam, China and the Philippines.
Multipurpose mobile starch extractor The perishable nature of tropical tuber crops and the difficulties in long-distance transport, storage, and marketing constitute major problems for farmers. In order to overcome this problem, a machine for in situ production of starch and value addition near the farm site is developed.	**Cassava chipping machine** The shorter shelf life of cassava tubers (up to 7 days in ambient conditions) limits their marketability and prevents farmers from transporting them to long-distance markets to obtain good prices. The cassava chipping machine helps the farmers to extend the shelf life of cassava tubers in dried form.
Biofortified Varieties of Tuber Crops Orange-fleshed sweet potato varieties contain Beta-carotene, a precursor of Vitamin A Low cost and natural way of combating Vitamin A deficiency.	**Others** CTCRI Bioformulations with Pesticidal Action.

CSIR–National Institute for Interdisciplinary Science and Technology (NIIST)
Industrial Estate P.O, Pappanamcode, Thiruvananthapuram, Kerala 695019

Ready to Cook tubers, starchy vegetables and pulse

The process is for making ready to cook products from regional origin. There are many regional crops which have healthy carbohydrates, micronutrients and phytochemicals which are popular among people of respective regions. But when we look at the consumption pattern of the present generation, refined foods have taken a dominant role leading to protein and micronutrient deficiencies. These products meet the nutrient requirement of consumers with less time investment in the kitchen.

Value added syrups from Palm neera and Coconut neera

Syrups prepared from saps of coconut and palmyra palm can be used for sweetening coffee, as toppings and can be a healthy substitute for ghur prepared from the sap of the palm and coconut tree. The same process can be adopted for making syrup from sugar cane juice and the syrup made can be a substitute for cane juice jaggery which is prepared by adding various additives for clarification and solidification. Our products are prepared with no preservatives.

Modular onsite wastewater treatment Cum resource recovery unit This is a modular unit that can treat organic rich wastewater from small establishments and recover reuse quality water and bioenergy. A patent has already been filed for this technology. Field units are working. Technology already licensed to one MSME for field implementation. Small establishments like canteens, restaurants, agro based MSMEs, community halls, etc.	**Biodegradable cutleries, cups, glass and plates from wheat barn, sugar cane bagasse, rice husk, fruit peels and pineapple leaves**  The developed product is shelf stable up to the period of 10 to 12 months and heat resistant upto the temperature of 100 0 C. It is found to be ideal for replacing the single use plastics and also the above said technology have been licensed and transferred to two MSME's for commercial production. It can be used in classic ovens/ microwave ovens. Benefits are Low cost, single usable, easily biodegradable and good water retention.
Fresh Ginger processing Technology Processing right from harvesting which is ideal for climatic conditions of North East States Fresh spice processing has the advantage of high-quality products with a fresh aroma and better yield Mechanical drying of spices operations offers processing during rainy seasons Employment opportunities for rural population and income generation to farmers The process is green and no usage of toxic chemicals. Features are • Yield of oil 30 % more than the conventional process. • Processing time reduced to 4 hours instead of 18 hours. • Reduced energy consumption. • Processing costs around 12 % of the total cost of production. • Superior quality of flavor.	**Process for development of weather resistant coir** This material can be used for construction of roads, protection of sea/river shores with improved strength and without frequent replacements in soil erosion prevention. The cost of treatment is only 10% of the cost of the geotextile.

Biodegradable Lignocellulose Fibre-based Mulching Mats and Sheets for Modern Farming Mulching is a covering, usually made of petroleum-based plastics, spread on the ground around plants to prevent excessive evaporation or erosion, inhibit weed growth, enrich the soil, support drip- irrigation, etc. Process know-how for the fabrication of thinner, flexible and rollable lignocellulosic fiber-based mulching mats using bio-based polymer binder for better crop growth. These mulching mats are biodegradable and eco-friendly substitutes to single-use plastic mulching films. A pilot-scale facility for the demonstration and fabrication of biodegradable mulching mats and sheets is available.	**Process for preparation of White Pepper** "White pepper", the de-skinned black or fresh pepper, is the most valued form of pepper (almost double the value of black pepper). Common method for making white pepper is traditional retting, which affects the product quality significantly. The innovative clean bioprocess from CSIR-NIIST helps in fast and bulk production of white pepper without losing its spicy principles. The process is designed to cleave the pectin molecular bonding between the skin and oil glands of the pepper kernel by the action of enzymes produced in-situ. This is facilitated in tanks by circulating liquid from a reservoir of microbial culture grown on degraded pepper skin medium. This bioprocess completes skin removal in 2 days for green and 4 days for black pepper under designed conditions.
Ready to Cook tubers, starchy vegetables and pulse The process is for making ready to cook products from regional origin. There are many regional crops which have healthy carbohydrates, micronutrients and phytochemicals which are popular among people of respective regions. But when we look at the consumption pattern of the present generation, refined foods have taken a dominant role leading to protein and micronutrient deficiencies. These products can meet the nutrient requirement of consumers with less time investment in the kitchen. Currently consumers shy away from elaborate cooking because of time and hence the availability of such products can cater to the need for healthy eating.	**Bio filter: Technology for Industrial Odor Control** Biofilter technology developed by CSIR-NIIST is unique for treatment of odorous gaseous streams. CSIR-NIIST undertakes careful study of customer requirements before designing the odor control system. NIIST Biofilter technology has been implemented in various industries i.e shrimp feed plants, fish meal plants, gelatine factory, bone meal factory, composting plant etc. NIIST high-performance, low-cost BIOFILTER MEDIA (US Pat. 6,696,284) has special characteristics like moisture, provides support of growth, provides nutrients, provides neutralizing agents, low pressure drop, high water holding capacity, enables mass transfer of contaminants from air to microbes, green and eco-friendly material, long life.

Kerala University of Fisheries and Ocean Studies (KUFOS)
Panangad Road, Madavana, Junction, Kochi, Kerala 682506

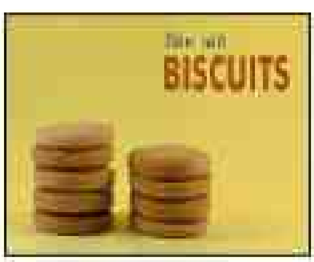
Feed for tilapia

The product was developed by replacing 50% of soybean meal with Cashew Nut Meal (CNM) in the diet of tilapia. This is an all plant ingredient based feed. CNM is locally available at a much cheaper price than SBM. CNM appears to be a suitable ingredient at 40% incorporation level along with 20% soya bean meal in the diet of tilapia. In addition, CNM also exhibited the lowest economic conversion ratio in this replacement level.

Marine Fiber rich Bread

Developed fiber-rich bread using carrageenan was prepared from osmotically-dehydrated snake-Gourd.

Marine Fiber rich Biscuit



Biscuits incorporated with marine fiber carrageenan was developed and found to have better nutritional characteristics, antioxidative properties and consumer acceptability. Incorporation of 6% carrageenan was found to be ideal for improved fiber content and consumer acceptability. The biscuits are packed in polypropylene cups and exhibited shelf life of the 56 days at room temperature.

Multi millet cookies

Incorporation of Palm jaggery the natural sweetner as well as multi-millet flours helps to improve the health benefits of cookies with very low glycemic index.
High levels of protein, fiber and various minerals. ragi and sorghum are gluten free. Curcumin is a natural antioxidant make the product capable to diminish effect of lifestyle induced oxidative stress. Carrageenan is a marine algae polysaccahride also improves the fiber content.

Fish gelatin based marshmallow

Fish gelatin, a multifunctional ingredient possesses an excellent whipping properties, used for the formation of foam in marshmallow. Marshmallow, an aerated confectionery product is very popular among kids because of

Fish gelatin based sugar coated gummy

Fish gelatin is a derivative product from the hydrolysis of collagen contained in the bones and skin of animals. It has a thermo-reversible gel formation ability and "melt in mouth texture", thus it acts as an active agents in the formulation of jelly-based confectioneries. Sugar coated gummy is a fish gelatin based chewable sweet with food acids such as citric

unique textural properties. The incorporation of gelatin provides a spongy structure and enhances the protein content of marshmallow.	acid or mallic acid in order to give a tart flavour. **Seaweed Pasta**  Application of seaweed based nutritional food is a viable current alternative to overcome this since seaweed cuisines are considered as highly valuable foodstuff around the globe, because of their nutritional importance. Pasta is considered a healthy food being relatively low in fat, high in carbohydrate, and with good protein content. Nutritional improvement of pasta mainly involves increasing protein and dietary fiber content and the fortification with vitamins and minerals. The addition of abundantly available seaweed as an ingredient in wheat pasta will enhance the nutritional profile.

Other technologies from KUFOS

1. Backyard Hatchery Technology for Karimeen
2. Backyard Hatchery Technology for the Olive Barb
3. Cashew nut waste based feed for tilapia
4. Farm made feed for Indian Major Carps (IMCs)
5. Insect based feed for *Pangasianodon hypophthalmus* (Pangasius)
6. Seed Production of Malabar Dwarf Puffer (*Carinotetraodon travancoricus*)
7. Ultra Violet Assisted Vertical Re-circulatory Depuration System
8. Battered and breaded products from farmed Basa (*Pangasius hypophthalmus*)
9. Calcium incorporated gluten-free cookies
10. Fish based pasta and noodles
11. Fish jerky
12. Fish soup powder Mola carplet
13. Fish gelatine based jelly
14. Fish gelatine based Marshmallow
15. Shrimp cracker
16. Seaweed pasta
17. Fish gelatine based Sugar coated gummy
18. Bilimbi syrup
19. Fibre rich biscuits
20. Diet Chocolate
21. Jackfruit leather
22. Marine fibre rich bread
23. Multi millet cookies
24. Nutrient bar
25. Thermally processed ready-to-eat tomato paste

ICAR- Sugarcane Breeding Institute Research Centre

Civil Station, John Mill Rd, near KIMST hospital back side entry, Talap, Kannur, Kerala 670002

LIQUID JAGGERY PROCESSING WITHOUT ANY CHEMICAL ADDITIVES

In the present method of liquid jaggery making chemical additives are not added at any stage of preparation. Heating facilitates the separation of dissolved colloidal albumins, wax and gum in the form dark coloured waxy coagulated matter floating on the surface. The liquid jaggery technology can be made with a hassle-free process without any chemical additives like clarificants, coloring agents and preservatives thereby ensuring the production of a healthy sweetener. This can also form an adulterant free healthy raw material for various confectionery items and in Ayurvedic Medicines.

POWDER JAGGERY PROCESSING FROM SUGARCANE JUICE WITH ORGANIC CLARIFICANTS

Powder jaggery preparation is a unique process by reducing the water content and obtains the product in powder form. The present method of powder jaggery preparation was standardized to get good uniform powder, better clarity, colour, taste and enhanced storability.

FREEZE PRESERVATION OF SUGARCANE JUICE WITH NATURAL ADDITIVES

The sugarcane juice changes its colour and taste soon after extraction and the shelf life is a concern. Within a short period, it completely gets spoiled due to polyphenol oxidation and microbial activity. The present method was standardized by quick freezing of the juice and adding the organic products which prevent polyphenol oxidation and avoid spoiling under storage. The addition of herbs of medicinal value can be a choice across age groups.

ICE CREAM PREPARATION BASED ON SUGARCANE SYRUP AND POWDER-JAGGERY

ICAR - Sugarcane Breeding Institute developed a technology to prepare ice cream from sugarcane syrup/ powder jaggery and sugarcane juice replacing the white sugar with a healthier form of sugar. Replacing white sugar with sugarcane syrup /powder jaggery and or sugarcane juice can enrich the ice creams with the natural flavour of sugarcane and the nutritional benefits of the sugarcane as such.

PRODUCTION OF SUGARCANE WINE WITH DIFFERENT FRUIT BLENDS

The sugarcane juice wine prepared independently lacks the aroma, color and was having low alcohol content<10%. Blending different fruit pulps with sugarcane juice is the novel step in the current study to bring the optimum alcohol content. The locally available and quickly perishable fruits were studied to ensure their availability in plenty. Preparing grape and other fruit-based wine requires the addition of white sugar which is not considered healthy sugar. Hence, a sugarcane juice-based wine will have better acceptability, flavor and color when we blend with different fruit pulps.

POWDER JAGGERY-BASED FORTIFIED COOKIES AND CAKES

CANE JAM PRODUCTION FROM SUGARCANE JUICE

KERALA AGRICULTURAL UNIVERSITY
VELLANIKKARA

1. Recycling of human hair waste into liquid fertilizer
2. Banana bunch covering device
3. Vertical Axial Flow Propeller Pump
4. Pineapple harvester attachment to backpack type brush cutter
5. Microbial Consortium for Aerobic Composting
   a) Liquid Formulation b) Coir pith base formulation
6. Novel Puttupodi
7. VCO based face cream
8. VCO based mouth wash
9. VCO based hair cream
10. Dehydrated Jackfruit Flakes
11. Osmo dehydrated ripe flakes
12. Seed flour for cattle feed
13. Osmodehydrated products for jack, pineapple, mango
14. Banana flour-based ethnic (Ragi, Njavara and Banana alone) mix
15. Extruded RTE snack from starch based food products
16. Bed former suitable Minitractor
17. KAU Bed Former attachable to tractor
18. Bioreactor for energy conversion of organic effluents
19. Motorized cocoa pod breaker
20. Processing protocol for Garcinia Powder
21. Balck Pepper Decorticator
22. Designing low-cost organic fertigation system for homestead vegetable production
23. Rotary Banana Slicer
24. Process Protocol for Banana Flour Production
25. Retort pouch packaged thermally processed tender jackfruit
26. Process protocol for kokum juice powder
27. Wax Applicator
28. RTC tuber fortified cold extruded pasta/ noodles products
29. RTE hot extruded expanded products
30. Jackfruit bulb slicer
31. Vacuum frying system for fruits and vegtables
32. Nutrient enriched spray dried Banana Pseudo stem juice powder
33. Tractor operated ginger planter
34. Coconut wood preservation - inorganic treatment
35. Spray dried coconut toddy powder
36. Spray dried neera powder
37. Coconut milk based ready-to serve
38. Jamun Nectar

ICAR-Indian Institute of Spices Research
(Indian Council of Agricultural Research)
Kozhikode-673012, Kerala, India

ICAR - INDIAN INSTITUTE OF SPICES RESEARCH

7RXV+25M, ISR Road, Moozhikkal - Parambil Bazar Rd, Vellimadukunnu, Kozhikode, Kerala 673012

Sl. No.	Technology*	Existing License fee (Rs)	License term	Royalty
1	A novel method of storing and delivering PGPR/ microbes through biocapsule.	10 Lakhs	5	2%
2	A seed coating composition and process for its preparation.	5.0 Lakhs	5	2%
3	*Trichoderma harzianum*, a biocontrol agent against *Phytophthora* foot rot-Talc formulation.	2.0 Lakhs	5	2%
4	*Pochonia chlamydosporia*, a biocontrol agent against nematodes- Talc formulation.	2.0 Lakhs	5	2%
5	*Pochonia chlamydosporia a biocontrol agent against nematodes*- liquid formulation.	3 .0 Lakhs	5	2%
6	PGPR microbial consortium for growth promotion in black pepper	1.0 Lakhs	5	2%
7	PGPR talc formation for seed spices	40,000/-	5	2%
8	PGPR talc formulation for ginger	60,000/-	5	2%

9	A micronutrient composition for ginger and a process for its preparation (for soils with pH above seven)	2.50 Lakhs	5	3%
10	A micronutrient composition for ginger and a process for its preparation (for soils with pH below seven)	2.50 Lakhs	5	3%
11	A micronutrient composition for turmeric and a process for its preparation (for soils with pH above seven)	2.50 Lakhs	5	3%
12	A micronutrient composition for turmeric and a process for its preparation (for soils with pH below seven)	2.50 Lakhs	5	3%
13	A micronutrient composition for black pepper and a process for its preparation	3.00 Lakhs	5	3%
14	A micronutrient composition for cardamom and a process for its preparation	3.00 Lakhs	5	3%
15	Golden Milk Mix	2.00 Lakhs	5	3%

* Reduction of Rs. 50000/- applicable on each technology (Sl. No 9-14 only) when more than one technology is availed at a time.

CENTRE FOR DEVELOPMENT OF ADVANCED COMPUTING

P.B.NO:6520, Vellayambalam, Thiruvananthapuram - 695033, Kerala, India

CerviSCAN CerviSCAN is an indigenised System for screening of Cervical Cancer using methodologies of Artificial Intelligence and robotics which makes population screening of Cervical Cancer Feasible. It has the following features 1) DIGISMEAR- AI Powered digital Cytology 2) RotoStainer- Robotic system for Slight staining 3) KytoSpin- Cyto Centrifuge 4) MFT- Indigenised Monolayer Smear Preparation	**MaxSim** Maxillo Facial Surgery Planning and Simulation System is a tool for diagnosis, Treatment Planning, and evaluation of maxillo-facial treatment result. **SRISTI** Software Reconstruction and Implant Synthesis form Topographic Images (SRISTI) is a software System for 3D visualization of CT and MRI images and creating physical models of Anatomical Parts and prosthetic implants through Rapid prototyping process. It provides a complete set of tools for visualizing 3D volume image data from th 2D Images captured to CT/MRI scanning there by decreasing the complexity in Image visualization.
TARANG TARANG is a feature rich state-of-the-art Digital Programmable hearing Aid which uses Sophisticated Digital Signal Processing techniques based on the indigenously developed Application Specific Integrated Circuits(ASIC) named NAADA, offering superior and stable characteristics over a wide dynamic range.	**ViEW** Virtual Endoscopy Visualization Software (ViEW) is jointly developed by CDAC, Thiruvananthapuram and Medical College Hospital Thiruvananthapuram incorporates the method for 3D reconstruction of Colon Structures CT imaged and provides interactive visualization and navigation through reconstructed colon.
CoSiCoSt-Env The Composite Signal Control Strategy (CoSiCoSt) optimizes a weighted combination of delay and number of stops in real-time. CoSiCoSt is designed to cater to the typical Indian driving and traffic conditions such as poor lane discipline and high heterogeneity.	**CoSMiC** Common SMart iot Connective is a middle ware software that is intended for deployment between an IOT Field device and an application software to provide standard based deployment of IOT/M2M ecosystem to ensure interoperability, security, data sharing and multi-vendor deployments of different applications in Intelligent Transportation system.

CUTE/WiTrac C-DAC Urban Traffic Control Equipment CUTE is a road traffic signal controller having features to perform as pre-timed controller or vehicle actuated controller at isolated intersections or part of a coordinated signal control system in pre-timed or vehicle actuated mode of operation or local level control of an Adaptive Traffic Control System (ATCS). CUTE is the latest edition of C-DAC traffic signal controllers after the UTCS and WiTraC.	**Emergency Service Vechicle Priority System(EmSerV)** The C-DAC Emergency Service Vehicle Priority System, EmSerV, has two modules (1) a controller unit installed inside the traffic signal controller and (2) the Vehicle Mount Unit (VMU) kept near the windshield of the Emergency Service Vehicle (ESV). EmSerV is implemented on geo-fencing technique using GPS coordinates. During an emergency trip the VMU broadcasts GPS coordinates (latitude and longitude) of the vehicle and its headway in every 500mSec using the 433MHz transmitter.
Pedestrian Safety Enhancement Controller(PeSCo) Presently available pedestrian signal controllers, in general, are designed for the walking speed of normal pedestrians. But, these days, a considerable number of 'divyang' (persons with disability) also use the motorway. The C-DAC developed Pedestrian Safety Enhancement Controller, PeSCo, is capable of identifying pedestrian demands through various input devices such as Pushbutton Switch, RFID and Ultrasonic Sensor for providing them sufficient crossing time based on the input device detected.	**TraMM-EnV** TraMM-EnV is one of the major building blocks of Advanced Traffic Management System (ATMS). This software tool is used to monitor and manage a network of C-DAC Adaptive Traffic signal controllers installed at different junctions in a City/Zone from the ATMS Command and Control Centre. TraMM-EnV incorporates user friendly Human Machine Interface, Dashboard to configure, visualize, analyze real traffic patterns and control traffic signals remotely.
FSS Mini The Miniature model of Full spectrum simulator is an ingeniously developed system that provides both offline and real time simulation capabilities at an affordable cost, easily configurable for custom applications. A Miniature version is configured as an educational package and small system simulation.	**GISPV** Grid Interactive Solar Photovoltaic export the power produced by the SPV array to the utility grid ensuring high power quality. Major components of the system are SPV array, Power conditioning unit, output filter and Interface Transformer. The PCU comprised of modular power converter hardware and its operation will be controlled through digital controller hardware platform built with advanced Digital Signal Processor and Field Programmable Gate Array.
HSRPEC High Speed Reconfigurable Power Electronics Controller is a reconfigurable controller architecture that can replace the conventional embedded micro controller/ Digital signal processor based controller design to real time control and monitoring of Power electronics system.	**MicroGrid for Remote Villages** Microgrid is a localized group of interconnected energy resources and loads which can either operate as an off grid system or in a grid connected environment. Off-grid microgrid is an idyllic solution for providing reliable power for rural areas where central grin is not a choice both geologically and economically.

SMART METER Energy Meters based on Indian Standards and suitable for Advanced Metering Infrastructure. Compatible with Smart Grid Communication Technologies and supports Distributed Generation.	**STATCOM** Static Compensators for Mitigation of Power quality Issues, are inverter based solutions used for compensation of reactive power, Harmonic current, unbalanced currents, and neutral currents caused by reactive and non-linear loads. When compared to conventional power quality solutions, STATCOMs features excellent dynamic performance under fluctuating load conditions, increased life, better efficiency, in sensitiveness to Grid frequency etc.
ARPWD Acoustic Red Palm Weevil Detector is an IOT device which detects gnawing sound using vibration sensors, processes the signals to extract the presence of RPW and notifies the farmers through SMS. The detection status will be sent via LoRaWAN Communication and stored in a remote data repository which is made available through Web Portal.	**NIRCHHARI** NIRCHHARI is an ultrasonic sensor system for remote level monitoring applications such waterways/water reservoirs, irrigation channels, underground city sewers, un attended diesel tanks at remote sites, rivers etc. The sensor measures and communicates periodic data from each site for live monitoring for a long time. The sensor also trasmits immediate alerts for events such as river flood alert, sewer overflow conditions, empty or full diesel tank.
SHRUTHI SHRUTHI is an application software for customizing the TARANG Hearing aids. It includes the client module, Audiometric module and Fitting Module. The software is intended to be used by qualified audiologist/doctors to customize the hearing aid for a particular patient, according to his/her hearing characteristics.	**Underwater Drone** Under Water Drone is a tethered submersible, controlled by a pilot located on land or from a vessel in water. An umbilical tether cable contains a group of cables that carry electrical power, video, and data signals and forth between the operator and ROS.

CENTRE FOR MATERIALS FOR ELECTRONICS TECHNOLOGY
(Under Ministry of Electronics & Information Technology, Govt. of India)

CENTER FOR MATERIALS FOR ELECTRONICS TECHNOLOGY (C-MET)
J667+GR2, Athani P.O., SH22, Kerala 680581

Fibre Optic Probe for Portable Sensors (Gas or bio- sensors)/association Fiber optic probe with cost-effective custom-developed coating for portable sensing applications. Demonstrated for various gasses including ammonia and hydrogen along with appropriate sensing layers and optical instrumentation. Working in the near infrared region and can be extended to online monitoring and remote sensing.	**NTC Fast Response Thermal Sensors** NTC Chip thermistors are extensively used for accurate temperature measurement and control in automobiles, medical field and electronic appliances. Fast response Chip in glass thermal sensor, an alternative to glass sealed bead and epoxy coated chip NTC thermistors combine the best characteristics of both types of thermal sensors. C-MET has developed variety of chip thermistor compositions and different sizes of chip thermistors having fast response. C-MET has developed chip in glass thermal sensors having an operating range up to 45^{0}C.
Graphene supercapacitors: Energy storage devices Supercapacitors can be used in a wide range of energy capture and storage applications. The advantages of supercapacitors include high power capability, long life, wide thermal operating range, low weight, flexible packaging, and low maintenance. They have already established their use in the automobile sector for hybrid drives and regenerative braking systems.	**Portable Biosensor with Disposable Chip** Portable biosensor with disposable 1 in x 1 in chip for detection of food borne bacteria. Validated with Salmonella, E Coli O157:H7, Shigella, etc. Jointly developed by C-MET Thrissur and Rajiv Gandhi Centre for Biotechnology(RGCB),Thiruvananthapuram.
Solar Lanterns: Renewable energy C-MET offers technology of three different variants of solar lanterns. Solar lantern with Lithium ion battery which can give light up to 8 hours and will be fully charged in 3 hrs in good sunlight. FM radio and high power torch light is also included in the product as additional features. Another variant is solar lantern with hybrid supercapacitor having a working voltage of 3.7V and this variant has fast charging compared to battery. Next variant is	**Thermistor based digital thermometers** The offered clinical digital thermometers are quite handy and can be used to instantly monitor any abnormalities in body temperature. The digital thermometer is provided with a larger digital display which makes it convenient to monitor the temperature with better readability. The clinical digital thermometer developed at C-MET has qualified all the tests as per IS 15113:2002 standard conducted by an NABL accredited

battery and supercapacitor hybrid solar lanterns which have the best features of both battery and supercapacitor based solar lanterns.	laboratory. The product is RoHS compliant. Design of the clinical digital thermometer is protected by filing design registration.
Transparent Conducting lines C-MET has developed a cost-effective process for coating low resistance electrical lines on glass sheets without blocking light. Ideal solution for making invisible electrical contacts on transparent glass. For example, LEDs connected at one end of a glass sheet can be lighted by applying electric power at the other end, using the transparent lines. **Wearable device for breast cancer screening: Medical devices/ tech transfer** C-MET has developed simple wearable devices (thermal sensors embedded in specially designed inner wear) for screening breast cancer with which women are most comfortable, ensures privacy, is absolutely painless, portable, takes only around 30 minutes and can even be operated by ASHA workers. This is a low cost device and the performance not only matches with other standard known techniques, in certain cases, additional details could also be discerned. The device is most ideal for mass screening of women in urban and rural settings. Technology is offered with 2D or 3D analysis software.	**Transparent Heater: Anti-frost coating** C-MET has developed a transparent heater for defrosting applications. The heater film coated on a glass panel through cost-effective technology provides uniform heating on its surface by applying electric power and can be employed in cold environments and for defrosting and anti-icing applications. Such heater films find applications in freezers kept in supermarkets to maintain transparency, in glass shelves kept in restaurants for keeping food items warm until service, automobile industry, etc.

Sree Chitra Tirunal Institute for Medical Sciences and Technology, Trivandrum

An Institution of National Importance, Department of Science and Technology, Govt. of India
Thiruvananthapuram, Kerala

1.Chitosan based antioxidant polymeric wound dressings for controlled antibiotic delivery

The product is a chitosan-based sponge for treating chronic non-healing as well as infectious wounds. The product is designed to deliver anti-microbial drugs to the wound site directly. The drug can be loaded by the doctor as per the need by a simple mechanism.

2. Dura Substitute

Dural defect reconstruction is a significant problem in many cases of decompressed craniotomy and also following skull base endoscopic surgeries. These defects are associated with a wide variety of lesions requiring neurosurgical interventions in order to obtain the re-establishment of the atomic barrier and to prevent the CSF leakage. CSF leakage is a complication associated with other possible serious complications like meningitis, and often requires re-operation. Dura substitution is indicated for use as a temporary or permanent prosthesis for repair of duramater. Electro spun polycarbonate urethane is developed as a synthetic non-degradable dura substitute.

3. Intracranial Electrodes

Epilepsy is the third most commonly diagnosed neurological disorder. Epileptic seizures are the result of excessive and abnormal nerve cell activity in the certain regions of brain. The surgical removal of the epileptogenic area of brain is the most effective treatment available for epilepsy. Intracranial electrodes help localize the epileptogenic zone, accurately and precisely by electrophysiological monitoring. The device employs platinum electrodes with Platinum - Iridium alloy connection wires embedded in silicone rubber. The device has high SNR for accurate signal acquisition.

4. Chitra GelMA UVS Bioink

Chitra UVS GelMA is an optimized formulation (bioink) that has been modified for Ultra Violet protection. The dried form of Chitra UVS GelMA can be easily solubilized in a physiological solution or cell culture media to get desired concentration of polymer and can be directly added to the cells. Rapid photo-crosslinking requires photoinitiator. Photocrosslinking of Chitra GelMA-UVS with UV of 365 nm for less than two minutes results in a stable hydrogel that provides a non-cytotoxic 3D environment to the cells. The novel bioink formulation shields the cells from harmful effect of UV during photocrosslinking. Cells adhere, migrate, proliferate and differentiate within the hydrogel. Hence Chitra GelMA-UVS finds application as a bioink that ensures viability and functionality of 3D bioprinted construct.

The present formulation of Chitra UVS GelMA is modified with UV protective molecules. Thus, it provides a safer microenvironment that ensures maximum cell viability when the cell laden hydrogel is subjected to UV irradiation for the purpose of cross-linking.

5. Recombinant growth factor for enhanced wound healing

Larger wounds or underlying disease conditions like diabetes often compromise the healing process. Faster healing is also essential for reducing scar formation. Growth factors are known to enhance wound healing. In this innovation, we developed two recombinant growth factors TGF alpha and VEGF, and an alginate-based scaffold for application to the wound to faster the healing process.

The novel concept for enhanced wound healing is incorporating growth factors in wound dressings. The preclinical evaluation showed that the recombinant growth factors, Vascular Endothelial Growth Factor-D (VEGF) and Transforming Growth Factor-alpha (TGF alpha) entrapped in an alginate scaffold delivery system accelerated wound healing.

6. Implantable micro infusion pump

Active programmable device to automatically deliver bolus or basal quantity of drug to the region of interest within human body such as peritoneal cavity, spinal cord or other diseased regions.

Features

• Configuration: Programmable • Flow Regime: Variable • Discharge Regime: Bolus/Basal/Priming.

-- --